Abdul Azeem Shaik

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US Citizen | No Sponsorship Needed

EDUCATION

Georgia Institute of Technology
M.S. in Computer Science, Specialization : Machine Learning — *Expected Dec 2026*

GPA: 4.0/4.0 | Graduate Researcher, Applied AI for Cybersecurity – Professor Vijay Madisetti

BITS Pilani
B.E. in Electronics and Communication Engineering — *Graduated 2023*

Societies: Member - ACM and IEEE, President - Department of Professional Events

PUBLICATION

"AI-based Market Intelligence Systems for Farmer Collectives" – ACM JCSS, funded by Google AI for Social Good; field-deployed researcher in rural India with IIT Delhi building custom solutions for voice interactions, price predictions and weather forecasts.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Rust, C++
Frameworks & Tools: Node.js, FastAPI, React, SQLite, Docker, Git, AWS Cloud Infrastructure
AI & Machine Learning: AWS Certified AI Practitioner; Vector DBs (Qdrant), RAG, PyTorch, HuggingFace, LangGraph, MCP

EXPERIENCE

Georgia Tech | *Graduate Researcher, Applied AI for Cybersecurity* — Apr 2026 – Sep 2026

- Built multi-agentic AI systems with multi-turn communication for autonomous security incident triage, adding fault-tolerant quorum voting so agents stay reliable in autonomous Security Operation Centre (SOC) settings. Open-sourced as SOC Defender.
- Outperformed frontier LLMs on Microsoft's Excytin and OpenSec cyber threat investigation benchmarks; invited by faculty to TA *Modern Internet Research Methods: Agentic AI*, and recommended for AI security research with an industry defense partner.
- Built a RAG system with SecureBERT embeddings and agent knowledge bases from MITRE ATT&CK, D3FEND, CWE and Sigma rules; fine-tuned and deployed Qwen2.5 14B with LoRA for SecOps domain knowledge; built an XGBoost + HDBSCAN classifier to detect malicious injection in IT logs.
- Evaluated prompt injection resistance through adversarial test pipelines based on Wazuh schemas for load testing.

Shardeum
Raised $24M in funding at a $248.9M valuation. Early technical hire, helped launch mainnet and testnet.

Senior Software Engineer, Distributed Systems — Jul 2025 – Dec 2025

- Built storage architecture and multi-shard state synchronization for network-wide consistency on a distributed platform scaling to 80M+ transactions and 150K+ registrations.
- Built payment APIs at ∼30 ms P99 latency, an RPC reverse proxy with node discovery and load balancing, and checkpoint-aware rollback and crash recovery for fault-tolerant operations. Set up production monitoring and alerting on AWS using CloudWatch, Lambda and Jenkins CI/CD.
- Built an MCP server for AI Agents enabling JSON-RPC communication for payment workflows and AI-assisted financial analysis.
- Led the redesign of authorization verification for incoming transactions; helped leadership model the network's governance & incentive plan; also built the cross-network payment platform and UI in React, Node.js and TypeScript.

Software Engineer, Distributed Systems — Jul 2023 – Jul 2025

- Improved ECDSA address recovery latency 14× (10 ms → 700 μs) by rewriting the core logic in Rust; improved distributed networking latency 34% (82 ms → 54 ms) in the TCP-based async library; cut signing latency 29% by moving the crypto library to Rust-based `libsodium`; built JSON-RPC features with GraphQL data services, reducing response delay by 78%.
- Built a custom serialization library for faster, secure gossip-based P2P; upgraded the data distribution protocol with WebSocket and RPC support.
- Found and fixed critical bounty bugs covering the Proof of Quorum consensus algorithm (fault tolerance and distributed coordination), DDoS, MITM and replay attacks, preventing double-spend exploits and unauthorized fund transfers on the live network; improved lost-node detection and added penalties for malicious activity.

Researcher — Aug 2022 – Jul 2023

- Handling 10,000 req/sec, designed and built a horizontally scalable node-monitoring service for decentralized communication networks, tracking node health and network state across a distributed peer set without a central bottleneck.

Liberdus | *Core Open Source Contributor & Maintainer*

- Improved reliability and performance of a distributed network: brought transaction time to finality down by 85% by fixing deadlocks in the signing layer; cut historical data sync latency by 63% by adding gossip protocols to the sync pipeline; reduced heap memory usage by 50% via RPC cache optimization, stopping OOM crashes; fixed database contention between data collectors and writer services using BetterSQLite.
- Primary security owner: audited the execution layer and fixed critical CVEs; hardened REST APIs against DDoS across the signing infrastructure and RPC clients; used agentic AI workflows to speed up production incident response (MTTR).
- Built a quorum-based validator mechanism for reliable client-side communication alongside configuration enforcement across distributed nodes; wrote cryptographic libraries in Rust for identity verification and threshold signatures.
- Also built user-facing payment products end to end, including the Liberdus Bridge transaction dashboard and payment UI in React and Node.js.
- **Other Merged Fixes:** **Red Hat** (MAPT, multi-cloud provisioning tool); **Farcaster** (Snapchain, social media platform).